Storage VM administration

Cloud Volumes ONTAP

NetApp
March 28, 2024

This PDF was generated from https://docs.netapp.com/us-en/bluexp-cloud-volumes-ontap/azure/task-managing-svms.html on March 28, 2024. Always check docs.netapp.com for the latest.

Table of Contents

Storage VM administration . . . 1
Manage storage VMs in BlueXP . . . 1
Create data-serving storage VMs for Cloud Volumes ONTAP in Azure . . . 2
Set up SVM disaster recovery . . . 23

Storage VM administration

Manage storage VMs in BlueXP

A storage VM is a virtual machine running within ONTAP that provides storage and data services to your clients. You might know this as an *SVM* or a *vserver*. Cloud Volumes ONTAP is configured with one storage VM by default, but some configurations support additional storage VMs.

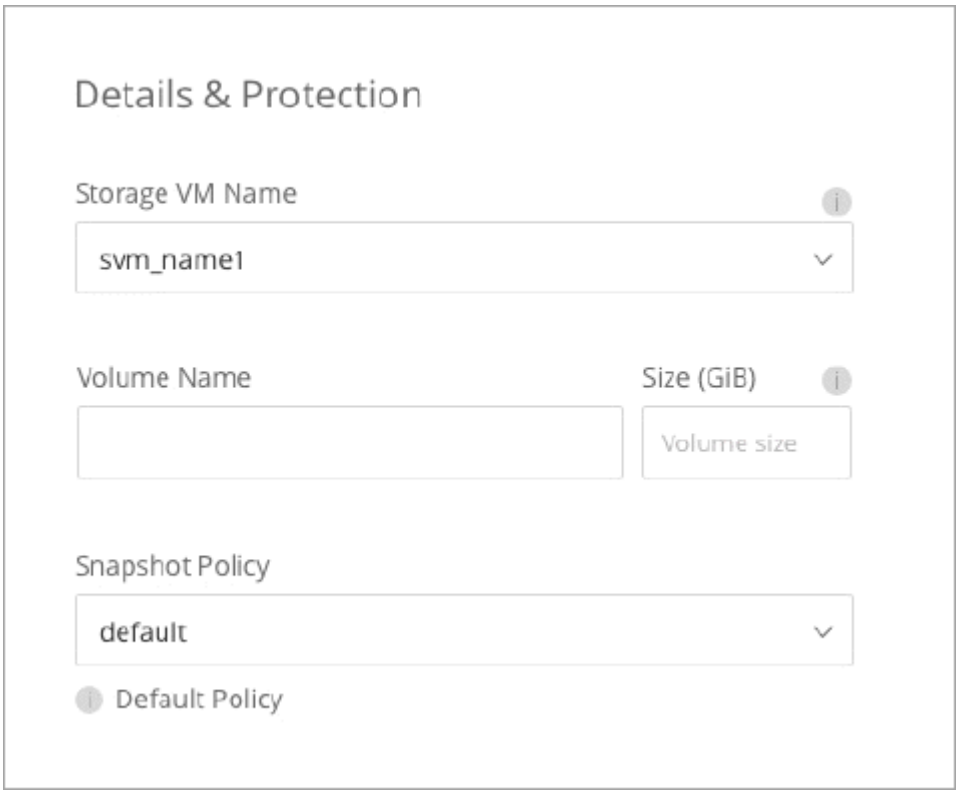
Supported number of storage VMs

Multiple storage VMs are supported with certain configurations. Go to the Cloud Volumes ONTAP Release Notes to verify the supported number of storage VMs for your version of Cloud Volumes ONTAP.

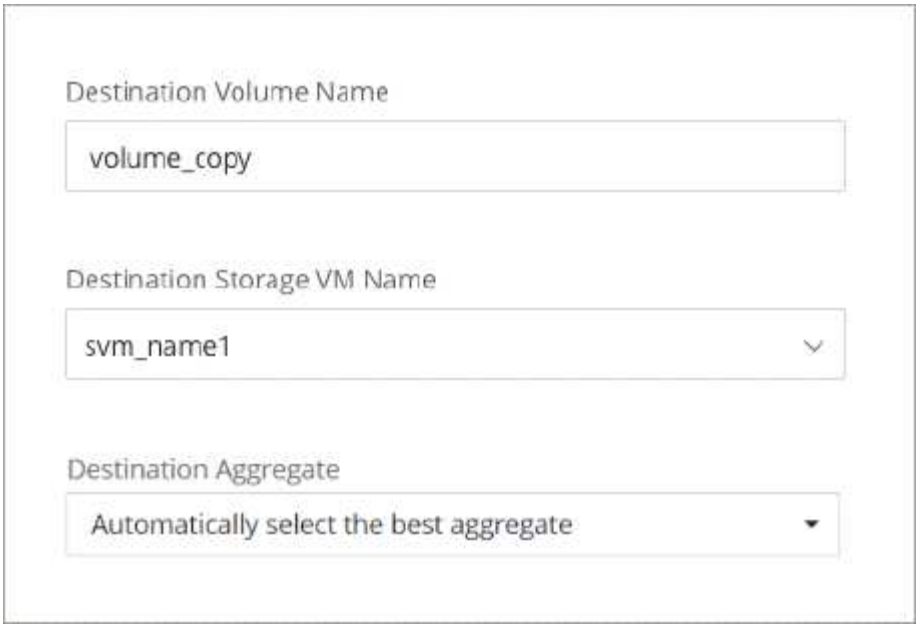
Work with multiple storage VMs

BlueXP supports any additional storage VMs that you create from System Manager or the CLI.

For example, the following image shows how you can choose a storage VM when you create a volume.



And the following image shows how you can choose a storage VM when replicating a volume to another system.



Modify the name of the default storage VM

BlueXP automatically names the single storage VM that it creates for Cloud Volumes ONTAP. From System Manager, CLI, or API, you can modify the name of the storage VM if you have strict naming standards. For example, you might want the name to match how you name the storage VMs for your ONTAP clusters.

Create data-serving storage VMs for Cloud Volumes ONTAP in Azure

A storage VM is a virtual machine running within ONTAP that provides storage and data services to your clients. You might know this as an *SVM* or a *vserver*. Cloud Volumes ONTAP is configured with one storage VM by default, but additional storage VMs are supported when running Cloud Volumes ONTAP in Azure.

To create additional data-serving storage VMs, you need to allocate IP addresses in Azure and then run ONTAP commands to create the storage VM and data LIFs.

> To perform additional NIC-related tasks, you can assign a network contributor role or custom role with appropriate permissions in Azure. For more information on these NIC-related permissions, see the Microsoft Azure documentation.

Supported number of storage VMs

Multiple storage VMs are supported with specific Cloud Volumes ONTAP configurations starting with the 9.9.0 release. Go to the Cloud Volumes ONTAP Release Notes to verify the supported number of storage VMs for your version of Cloud Volumes ONTAP.

All other Cloud Volumes ONTAP configurations support one data-serving storage VM and one destination storage VM used for disaster recovery. You can activate the destination storage VM for data access if there's an outage on the source storage VM.

Allocate IP addresses in Azure

You need to allocate IP addresses in Azure before you create a storage VM and allocate LIFs.

Single node system

IP addresses must be assigned to nic0 in Azure before you create a storage VM and allocate LIFs.

You'll need to create an IP address for data LIF access and another optional IP address for a storage VM (SVM) management LIF. This management LIF provides a connection to management tools like SnapCenter.

Steps

1. Log in to the Azure portal and open the **Virtual machine** service.
2. Click the name of the Cloud Volumes ONTAP VM.
3. Click **Networking**.
4. Click the name of the network interface for nic0.
5. Under **Settings**, click **IP configurations**.
6. Click **Add**.
7. Enter a name for the IP configuration, select **Dynamic**, and then click **OK**.
8. Click the name of the IP configuration that you just created, change the **Assignment** to **Static**, and click **Save**.

   It's best to use a static IP address because a static IP ensures that the IP address won't change, which can help to prevent unnecessary outages to your application.

If you want to create an SVM management LIF, repeat these steps to create an additional IP address.

After you finish

Copy the private IP addresses that you just created. You'll need to specify those IP addresses when you create LIFs for the new storage VM.

HA pair

How you allocate IP addresses for an HA pair depends on the storage protocol that you're using.

iSCSI

iSCSI IP addresses must be assigned to nic0 in Azure before you create a storage VM and allocate LIFs. IPs for iSCSI are assigned to nic0 and not the load balancer because iSCSI uses ALUA for failover.

You'll need to create the following IP addresses:

- One IP address for iSCSI data LIF access from node 1
- One IP address for iSCSI data LIF access from node 2
- An optional IP address for a storage VM (SVM) management LIF

  This management LIF provides a connection to management tools like SnapCenter.

Steps

1. Log in to the Azure portal and open the **Virtual machine** service.
2. Click the name of the Cloud Volumes ONTAP VM for node 1.
3. Click **Networking**.
4. Click the name of the network interface for nic0.
5. Under **Settings**, click **IP configurations**.
6. Click **Add**.
7. Enter a name for the IP configuration, select **Dynamic**, and then click **OK**.
8. Click the name of the IP configuration that you just created, change the **Assignment** to **Static**, and click **Save**.

   It's best to use a static IP address because a static IP ensures that the IP address won't change, which can help to prevent unnecessary outages to your application.

9. Repeat these steps on node 2.
10. If you want to create an SVM management LIF, repeat these steps on node 1.

NFS

IP addresses that you use for NFS are allocated in the load balancer so that the IP addresses can migrate to the other node in case failover events occur.

You'll need to create the following IP addresses:

- One IP address for NAS data LIF access from node 1
- One IP address for NAS data LIF access from node 2
- An optional IP address for a storage VM (SVM) management LIF

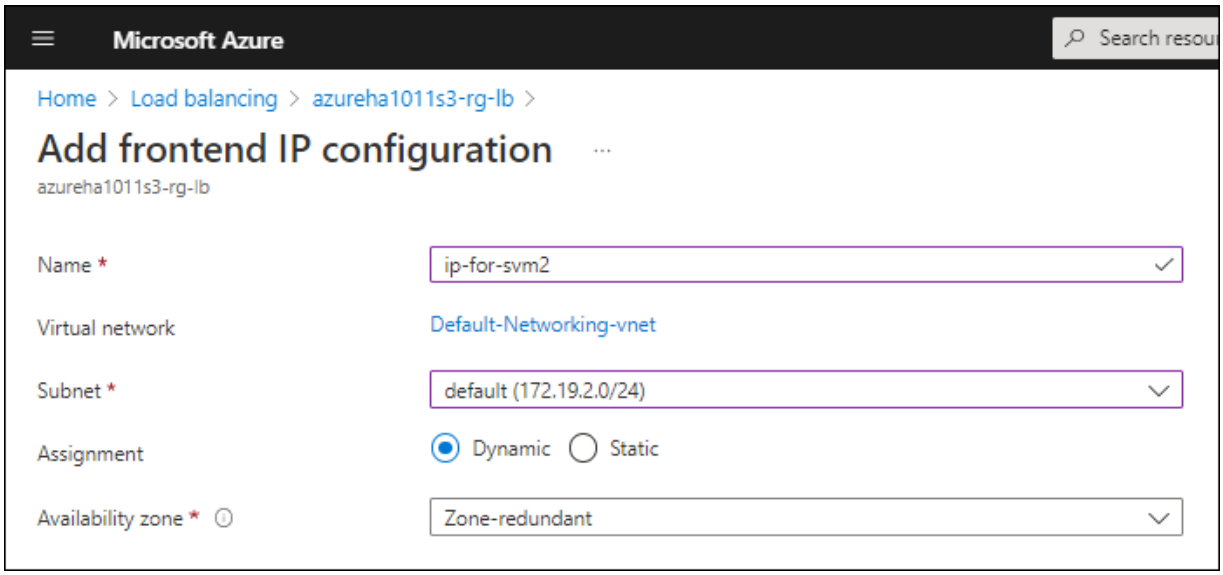
  The iSCSI LIFs are required for DNS communication. An iSCSI LIF is used for this purpose because it doesn't migrate on failover.

  This management LIF provides a connection to management tools like SnapCenter.

Steps

1. In the Azure portal, open the **Load balancers** service.

2. Click the name of the load balancer for the HA pair.
3. Create one frontend IP configuration for data LIF access from node 1, another for data LIF access from node 2, and another optional frontend IP for a storage VM (SVM) management LIF.
   a. Under **Settings**, click **Frontend IP configuration**.
   b. Click **Add**.
   c. Enter a name for the frontend IP, select the subnet for the Cloud Volumes ONTAP HA pair, leave **Dynamic** selected, and in regions with Availability Zones, leave **Zone-redundant** selected to ensure that the IP address remains available if a zone fails.

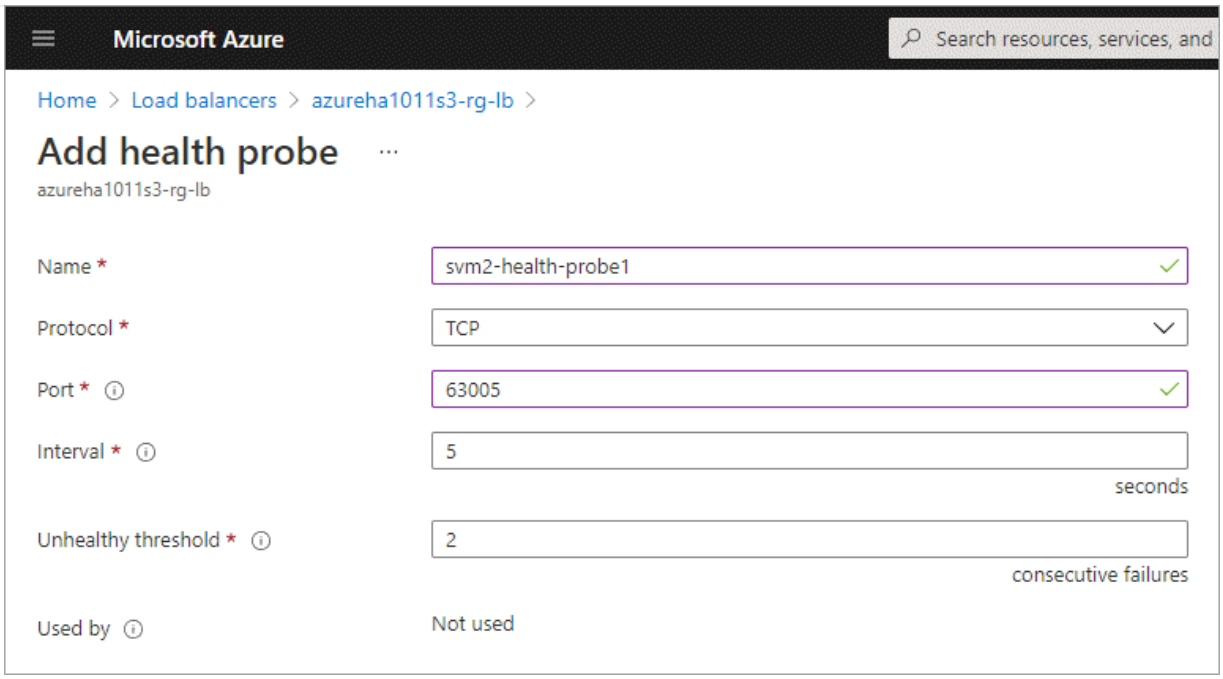


   d. Click the name of the frontend IP configuration that you just created, change the **Assignment** to **Static**, and click **Save**.

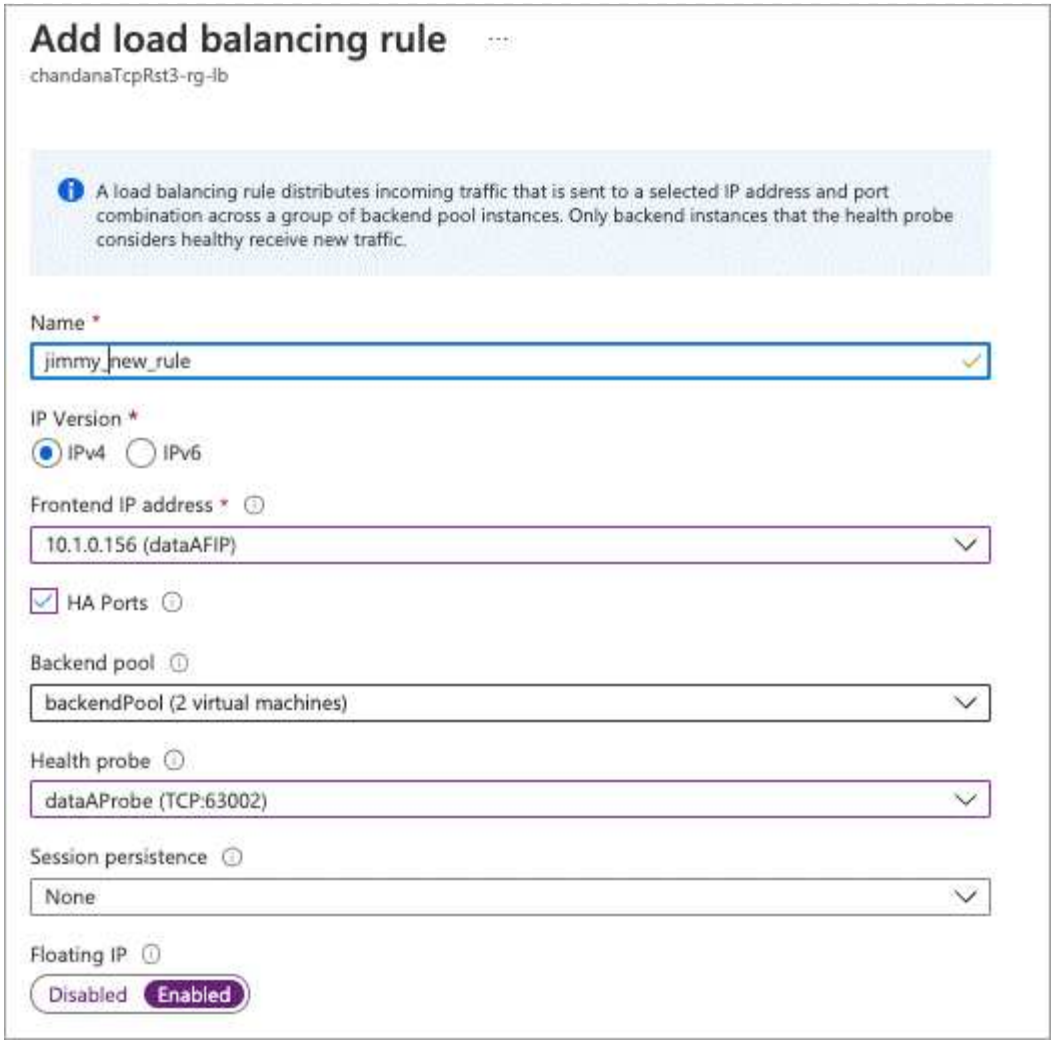
      It's best to use a static IP address because a static IP ensures that the IP address won't change, which can help to prevent unnecessary outages to your application.

4. Add a health probe for each frontend IP that you just created.
   a. Under the load balancer's **Settings**, click **Health probes**.
   b. Click **Add**.
   c. Enter a name for the health probe and enter a port number that's between 63005 and 65000. Keep the default values for the other fields.

      It's important that the port number is between 63005 and 65000. For example, if you are creating three health probes, you could enter probes that use the port numbers 63005, 63006, and 63007.



5. Create new load balancing rules for each frontend IP.
   a. Under the load balancer's **Settings**, click **Load balancing rules**.
   b. Click **Add** and enter the required information:
      - **Name**: Enter a name for the rule.
      - **IP Version**: Select **IPv4**.
      - **Frontend IP address**: Select one of the frontend IP addresses that you just created.
      - **HA Ports**: Enable this option.
      - **Backend pool**: Keep the default Backend pool that was already selected.
      - **Health probe**: Select the health probe that you created for the selected frontend IP.
      - **Session persistence**: Select **None**.
      - **Floating IP**: Select **Enabled**.



6. Ensure that the network security group rules for Cloud Volumes ONTAP allows the load balancer to send TCP probes for the health probes that were created in step 4 above. Note that this is allowed by default.

SMB

IP addresses that you use for SMB data are allocated in the load balancer so that the IP addresses can migrate to the other node in case failover events occur.

You'll need to create the following IP addresses in the load balancer:

- One IP address for NAS data LIF access from node 1
- One IP address for NAS data LIF access from node 2
- One IP address for an iSCSI LIF on node 1 in each VM's respective NIC0
- One IP address for an iSCSI LIF on node 2

  The iSCSI LIFs are required for DNS and SMB communication. An iSCSI LIF is used for this purpose because it doesn't migrate on failover.

- An optional IP address for a storage VM (SVM) management LIF

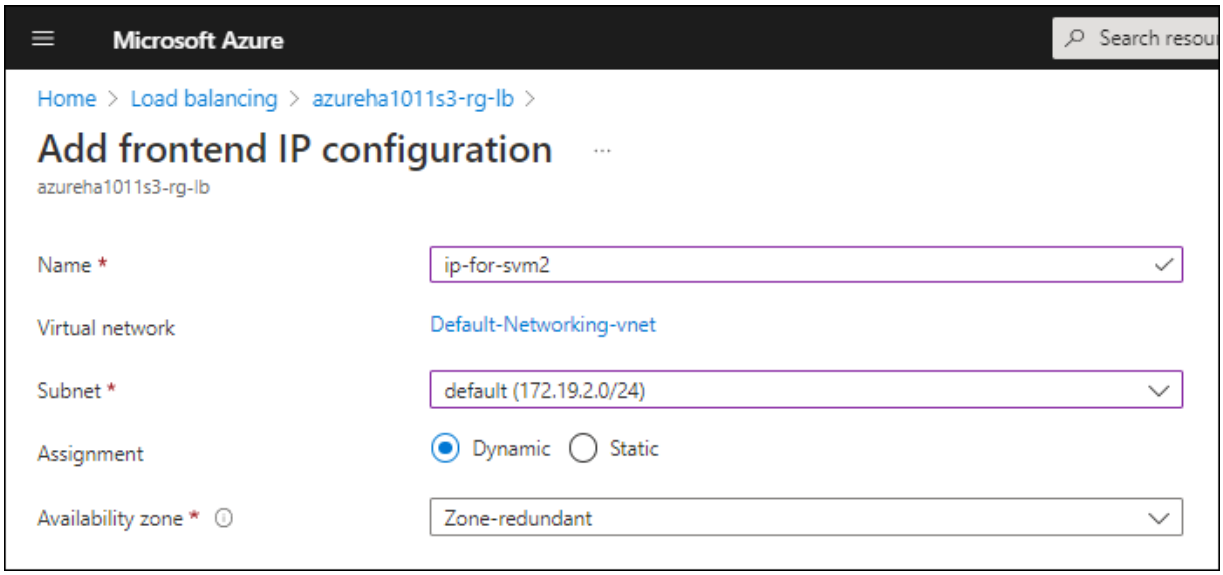
  This management LIF provides a connection to management tools like SnapCenter.

Steps

1. In the Azure portal, open the **Load balancers** service.
2. Click the name of the load balancer for the HA pair.
3. Create the required number of frontend IP configurations for the data and SVM LIFs only:

> A frontend IP should only be created under the NIC0 for each corresponding SVM. For more information on how to add the IP address to the SVM NIC0, see "Step 7 [hyperlink]"

   a. Under **Settings**, click **Frontend IP configuration**.
   b. Click **Add**.
   c. Enter a name for the frontend IP, select the subnet for the Cloud Volumes ONTAP HA pair, leave **Dynamic** selected, and in regions with Availability Zones, leave **Zone-redundant** selected to ensure that the IP address remains available if a zone fails.

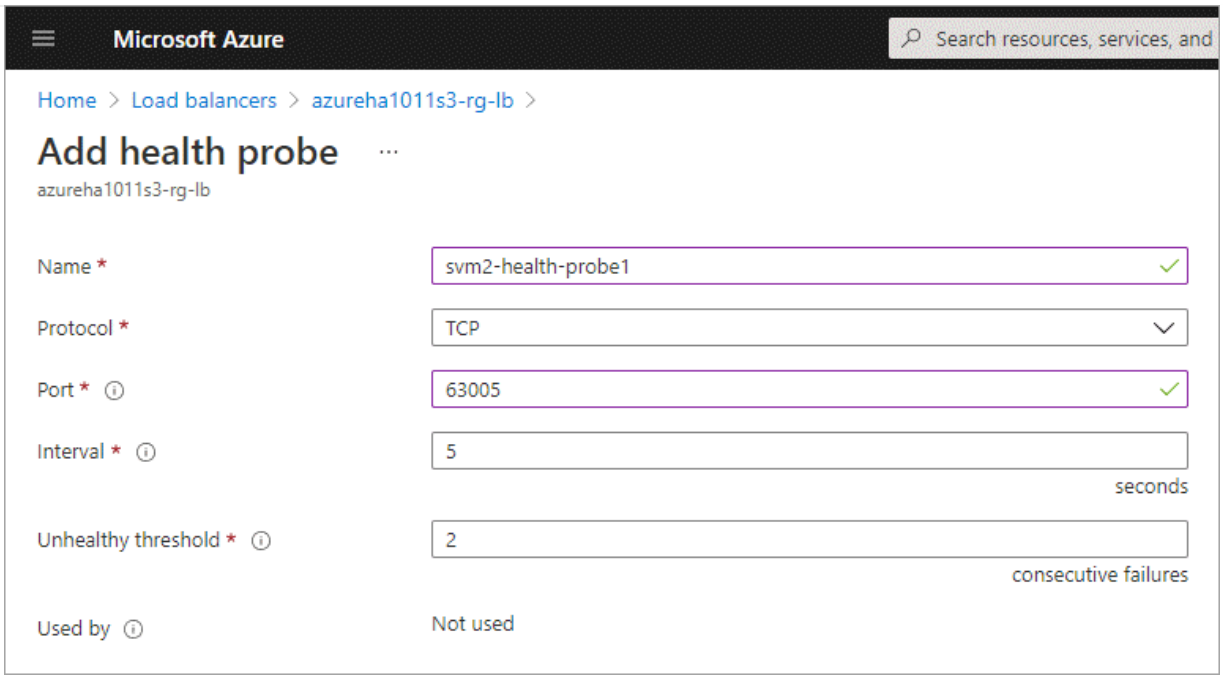


   d. Click the name of the frontend IP configuration that you just created, change the **Assignment** to **Static**, and click **Save**.

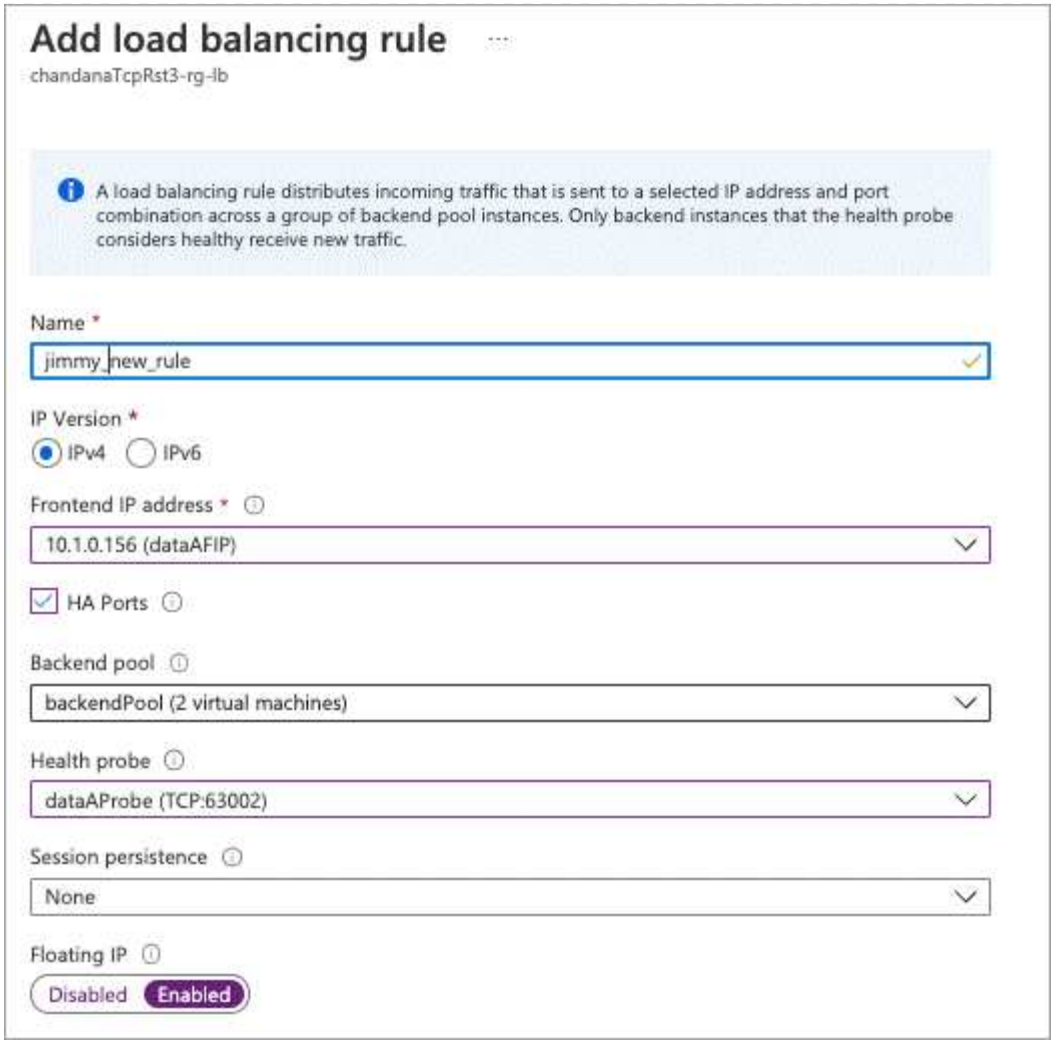
      It's best to use a static IP address because a static IP ensures that the IP address won't change, which can help to prevent unnecessary outages to your application.

4. Add a health probe for each frontend IP that you just created.
   a. Under the load balancer's **Settings**, click **Health probes**.
   b. Click **Add**.
   c. Enter a name for the health probe and enter a port number that's between 63005 and 65000. Keep the default values for the other fields.

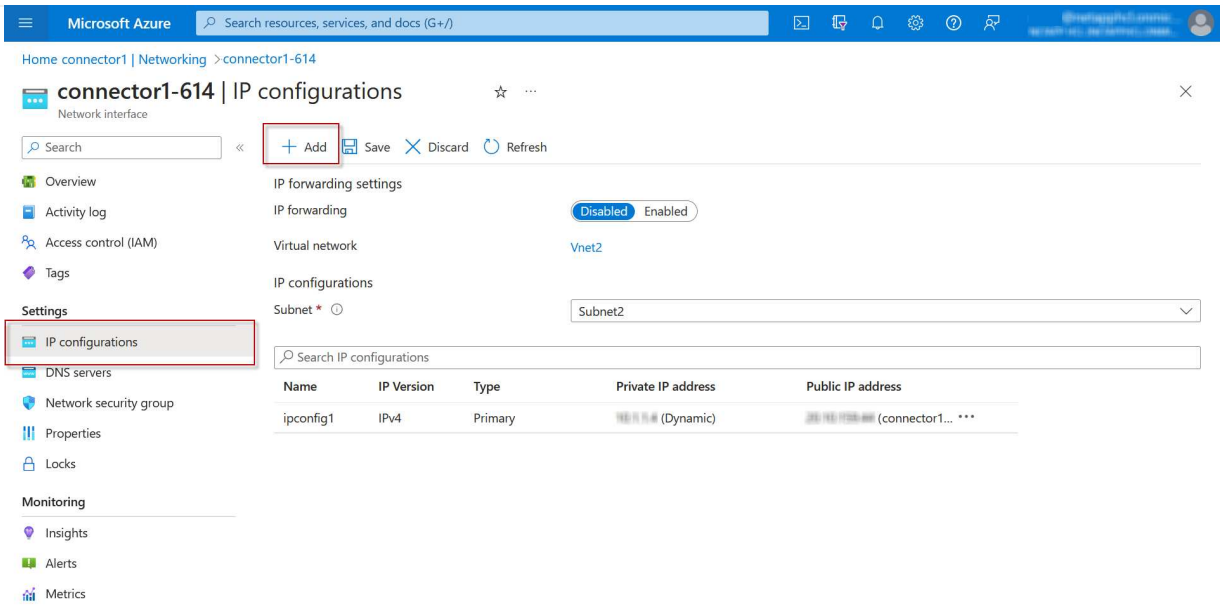
      It's important that the port number is between 63005 and 65000. For example, if you are creating three health probes, you could enter probes that use the port numbers 63005, 63006, and 63007.



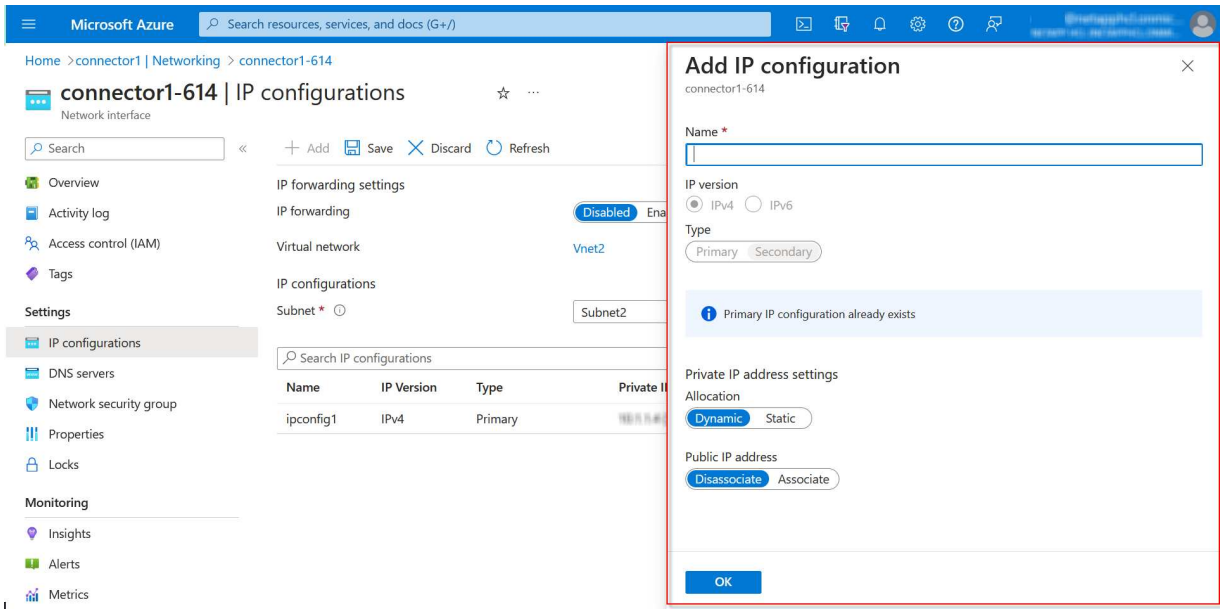
5. Create new load balancing rules for each frontend IP.
   a. Under the load balancer's **Settings**, click **Load balancing rules**.
   b. Click **Add** and enter the required information:
      - **Name**: Enter a name for the rule.
      - **IP Version**: Select **IPv4**.
      - **Frontend IP address**: Select one of the frontend IP addresses that you just created.
      - **HA Ports**: Enable this option.
      - **Backend pool**: Keep the default Backend pool that was already selected.
      - **Health probe**: Select the health probe that you created for the selected frontend IP.
      - **Session persistence**: Select **None**.
      - **Floating IP**: Select **Enabled**.



6. Ensure that the network security group rules for Cloud Volumes ONTAP allows the load balancer to send TCP probes for the health probes that were created in step 4 above. Note that this is allowed by default.
7. For iSCSI LIFs, add the IP address for NIC0.
   a. Click the name of the Cloud Volumes ONTAP VM.
   b. Click **Networking**.
   c. Click the name of the network interface for nic0.
   d. Under Settings, click **IP configurations**.
   e. Click **Add**.



f. Enter a name for the IP configuration, select Dynamic, and then click **OK**.



g. Click the name of the IP configuration that you just created, change the Assignment to Static, and click **Save**.

> It's best to use a static IP address because a static IP ensures that the IP address won't change, which can help to prevent unnecessary outages to your application.

After you finish

Copy the private IP addresses that you just created. You'll need to specify those IP addresses when you create LIFs for the new storage VM.

Create a storage VM and LIFs

After you allocate IP addresses in Azure, you can create a new storage VM on a single node system or on an

HA pair.

Single node system

How you create a storage VM and LIFs on a single node system depends on the storage protocol that you're using.

iSCSI

Follow these steps to create a new storage VM, along with the required LIFs.

Steps

1. Create the storage VM and a route to the storage VM.

```
vserver create -vserver <svm-name> -subtype default -rootvolume
<root-volume-name> -rootvolume-security-style unix
```

```
network route create -destination 0.0.0.0/0 -vserver <svm-name>
-gateway <ip-of-gateway-server>
```

2. Create a data LIF:

```
network interface create -vserver <svm-name> -home-port e0a -address
<iscsi-ip-address> -netmask-length <# of mask bits> -lif <lif-name>
-home-node <name-of-node1> -data-protocol iscsi
```

3. Optional: Create a storage VM management LIF.

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol none -address <svm-mgmt-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy system-defined -firewall-policy mgmt -home-port e0a
-auto-revert false -failover-group Default
```

4. Assign one or more aggregates to the storage VM.

```
vserver add-aggregates -vserver svm_2 -aggregates aggr1,aggr2
```

   This step is required because the new storage VM needs access to at least one aggregate before you can create volumes on the storage VM.

NFS

Follow these steps to create a new storage VM, along with the required LIFs.

Steps

1. Create the storage VM and a route to the storage VM.

```
vserver create -vserver <svm-name> -subtype default -rootvolume
<root-volume-name> -rootvolume-security-style unix
```

```
network route create -destination 0.0.0.0/0 -vserver <svm-name>
-gateway <ip-of-gateway-server>
```

2. Create a data LIF:

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol cifs,nfs -address <nas-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy disabled -firewall-policy data -home-port e0a -auto
-revert true -failover-group Default
```

3. Optional: Create a storage VM management LIF.

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol none -address <svm-mgmt-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy system-defined -firewall-policy mgmt -home-port e0a
-auto-revert false -failover-group Default
```

4. Assign one or more aggregates to the storage VM.

```
vserver add-aggregates -vserver svm_2 -aggregates aggr1,aggr2
```

This step is required because the new storage VM needs access to at least one aggregate before you can create volumes on the storage VM.

SMB

Follow these steps to create a new storage VM, along with the required LIFs.

Steps

1. Create the storage VM and a route to the storage VM.

```
vserver create -vserver <svm-name> -subtype default -rootvolume
<root-volume-name> -rootvolume-security-style unix
```

```
network route create -destination 0.0.0.0/0 -vserver <svm-name>
-gateway <ip-of-gateway-server>
```

2. Create a data LIF:

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol cifs,nfs -address <nas-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy disabled -firewall-policy data -home-port e0a -auto
-revert true -failover-group Default
```

3. Optional: Create a storage VM management LIF.

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol none -address <svm-mgmt-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy system-defined -firewall-policy mgmt -home-port e0a
-auto-revert false -failover-group Default
```

4. Assign one or more aggregates to the storage VM.

```
vserver add-aggregates -vserver svm_2 -aggregates aggr1,aggr2
```

   This step is required because the new storage VM needs access to at least one aggregate before you can create volumes on the storage VM.

HA pair

How you create a storage VM and LIFs on an HA pair depends on the storage protocol that you’re using.

iSCSI

Follow these steps to create a new storage VM, along with the required LIFs.

Steps

1. Create the storage VM and a route to the storage VM.

```
vserver create -vserver <svm-name> -subtype default -rootvolume
<root-volume-name> -rootvolume-security-style unix
```

```
network route create -destination 0.0.0.0/0 -vserver <svm-name>
-gateway <ip-of-gateway-server>
```

2. Create data LIFs:
   a. Use the following command to create an iSCSI LIF on node 1.

```
network interface create -vserver <svm-name> -home-port e0a
-address <iscsi-ip-address> -netmask-length <# of mask bits> -lif
<lif-name> -home-node <name-of-node1> -data-protocol iscsi
```

   b. Use the following command to create an iSCSI LIF on node 2.

```
network interface create -vserver <svm-name> -home-port e0a
-address <iscsi-ip-address> -netmask-length <# of mask bits> -lif
<lif-name> -home-node <name-of-node2> -data-protocol iscsi
```

3. Optional: Create a storage VM management LIF on node 1.

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol none -address <svm-mgmt-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy system-defined -firewall-policy mgmt -home-port e0a
-auto-revert false -failover-group Default
```

   This management LIF provides a connection to management tools like SnapCenter.

4. Assign one or more aggregates to the storage VM.

```
vserver add-aggregates -vserver svm_2 -aggregates aggr1,aggr2
```

   This step is required because the new storage VM needs access to at least one aggregate before you

can create volumes on the storage VM.

5. If you're running Cloud Volumes ONTAP 9.11.1 or later, modify the network service policies for the storage VM.

   a. Enter the following command to access advanced mode.

   ```
   ::> set adv -con off
   ```

   Modifying the services is required because it ensures that Cloud Volumes ONTAP can use the iSCSI LIF for outbound management connections.

   ```
   network interface service-policy remove-service -vserver <svm-
   name> -policy default-data-files -service data-fpolicy-client
   network interface service-policy remove-service -vserver <svm-
   name> -policy default-data-files -service management-ad-client
   network interface service-policy remove-service -vserver <svm-
   name> -policy default-data-files -service management-dns-client
   network interface service-policy remove-service -vserver <svm-
   name> -policy default-data-files -service management-ldap-client
   network interface service-policy remove-service -vserver <svm-
   name> -policy default-data-files -service management-nis-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-blocks -service data-fpolicy-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-blocks -service management-ad-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-blocks -service management-dns-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-blocks -service management-ldap-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-blocks -service management-nis-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-iscsi -service data-fpolicy-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-iscsi -service management-ad-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-iscsi -service management-dns-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-iscsi -service management-ldap-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-iscsi -service management-nis-client
   ```

NFS

Follow these steps to create a new storage VM, along with the required LIFs.

Steps

1. Create the storage VM and a route to the storage VM.

```
vserver create -vserver <svm-name> -subtype default -rootvolume
<root-volume-name> -rootvolume-security-style unix
```

```
network route create -destination 0.0.0.0/0 -vserver <svm-name>
-gateway <ip-of-gateway-server>
```

2. Create data LIFs:
   a. Use the following command to create a NAS LIF on node 1.

```
network interface create -vserver <svm-name> -lif <lif-name>
-role data -data-protocol cifs,nfs -address <nfs-cifs-ip-address>
-netmask-length <length> -home-node <name-of-node1> -status-admin
up -failover-policy system-defined -firewall-policy data -home
-port e0a -auto-revert true -failover-group Default -probe-port
<port-number-for-azure-health-probe1>
```

   b. Use the following command to create a NAS LIF on node 2.

```
network interface create -vserver <svm-name> -lif <lif-name>
-role data -data-protocol cifs,nfs -address <nfs-cifs-ip-address>
-netmask-length <length> -home-node <name-of-node2> -status-admin
up -failover-policy system-defined -firewall-policy data -home
-port e0a -auto-revert true -failover-group Default -probe-port
<port-number-for-azure-health-probe2>
```

3. Create iSCSI LIFs to provide DNS communication:
   a. Use the following command to create an iSCSI LIF on node 1.

```
network interface create -vserver <svm-name> -home-port e0a
-address <iscsi-ip-address> -netmask-length <# of mask bits> -lif
<lif-name> -home-node <name-of-node1> -data-protocol iscsi
```

   b. Use the following command to create an iSCSI LIF on node 2.

```
network interface create -vserver <svm-name> -home-port e0a
-address <iscsi-ip-address> -netmask-length <# of mask bits> -lif
<lif-name> -home-node <name-of-node2> -data-protocol iscsi
```

4. Optional: Create a storage VM management LIF on node 1.

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol none -address <svm-mgmt-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy system-defined -firewall-policy mgmt -home-port e0a
-auto-revert false -failover-group Default -probe-port <port-number-
for-azure-health-probe3>
```

   This management LIF provides a connection to management tools like SnapCenter.

5. Optional: Create a storage VM management LIF on node 1.

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol none -address <svm-mgmt-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy system-defined -firewall-policy mgmt -home-port e0a
-auto-revert false -failover-group Default -probe-port <port-number-
for-azure-health-probe3>
```

   This management LIF provides a connection to management tools like SnapCenter.

6. Assign one or more aggregates to the storage VM.

```
vserver add-aggregates -vserver svm_2 -aggregates aggr1,aggr2
```

   This step is required because the new storage VM needs access to at least one aggregate before you can create volumes on the storage VM.

7. If you’re running Cloud Volumes ONTAP 9.11.1 or later, modify the network service policies for the storage VM.
   a. Enter the following command to access advanced mode.

```
::> set adv -con off
```

      Modifying the services is required because it ensures that Cloud Volumes ONTAP can use the iSCSI LIF for outbound management connections.

```
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service data-fpolicy-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-ad-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-dns-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-ldap-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-nis-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service data-fpolicy-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-ad-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-dns-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-ldap-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-nis-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service data-fpolicy-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-ad-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-dns-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-ldap-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-nis-client
```

SMB

Follow these steps to create a new storage VM, along with the required LIFs.

Steps

1. Create the storage VM and a route to the storage VM.

```
vserver create -vserver <svm-name> -subtype default -rootvolume
<root-volume-name> -rootvolume-security-style unix
```

```
network route create -destination 0.0.0.0/0 -vserver <svm-name>
-gateway <ip-of-gateway-server>
```

2. Create NAS data LIFs:
   a. Use the following command to create a NAS LIF on node 1.

```
network interface create -vserver <svm-name> -lif <lif-name>
-role data -data-protocol cifs,nfs -address <nfs-cifs-ip-address>
-netmask-length <length> -home-node <name-of-node1> -status-admin
up -failover-policy system-defined -firewall-policy data -home
-port e0a -auto-revert true -failover-group Default -probe-port
<port-number-for-azure-health-probe1>
```

   b. Use the following command to create a NAS LIF on node 2.

```
network interface create -vserver <svm-name> -lif <lif-name>
-role data -data-protocol cifs,nfs -address <nfs-cifs-ip-address>
-netmask-length <length> -home-node <name-of-node2> -status-admin
up -failover-policy system-defined -firewall-policy data -home
-port e0a -auto-revert true -failover-group Default -probe-port
<port-number-for-azure-health-probe2>
```

3. Create iSCSI LIFs to provide DNS communication:
   a. Use the following command to create an iSCSI LIF on node 1.

```
network interface create -vserver <svm-name> -home-port e0a
-address <iscsi-ip-address> -netmask-length <# of mask bits> -lif
<lif-name> -home-node <name-of-node1> -data-protocol iscsi
```

   b. Use the following command to create an iSCSI LIF on node 2.

```
network interface create -vserver <svm-name> -home-port e0a
-address <iscsi-ip-address> -netmask-length <# of mask bits> -lif
<lif-name> -home-node <name-of-node2> -data-protocol iscsi
```

4. Optional: Create a storage VM management LIF on node 1.

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol none -address <svm-mgmt-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy system-defined -firewall-policy mgmt -home-port e0a
-auto-revert false -failover-group Default -probe-port <port-number-
for-azure-health-probe3>
```

This management LIF provides a connection to management tools like SnapCenter.

5. Assign one or more aggregates to the storage VM.

```
vserver add-aggregates -vserver svm_2 -aggregates aggr1,aggr2
```

This step is required because the new storage VM needs access to at least one aggregate before you can create volumes on the storage VM.

6. If you’re running Cloud Volumes ONTAP 9.11.1 or later, modify the network service policies for the storage VM.

   a. Enter the following command to access advanced mode.

```
::> set adv -con off
```

   Modifying the services is required because it ensures that Cloud Volumes ONTAP can use the iSCSI LIF for outbound management connections.

```
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service data-fpolicy-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-ad-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-dns-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-ldap-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-nis-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service data-fpolicy-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-ad-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-dns-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-ldap-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-nis-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service data-fpolicy-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-ad-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-dns-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-ldap-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-nis-client
```

What's next?

After you create a storage VM on an HA pair, it's best to wait 12 hours before you provision storage on that SVM. Starting with the Cloud Volumes ONTAP 9.10.1 release, BlueXP scans the settings for an HA pair's load balancer at a 12-hour interval. If there are new SVMs, BlueXP will enable a setting that provides shorter unplanned failover.

Set up SVM disaster recovery

BlueXP doesn't provide any setup or orchestration support for storage VM (SVM) disaster recovery. You must use System Manager or the CLI.

If you set up SnapMirror SVM replication between two Cloud Volumes ONTAP systems, the replication must be between two HA pair systems or two single node systems. You can't set up SnapMirror SVM replication between an HA pair and a single node system.

Refer to the following documents for CLI instructions.

- SVM Disaster Recovery Preparation Express Guide
- SVM Disaster Recovery Express Guide

Copyright information

Copyright © 2024 NetApp, Inc. All Rights Reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means—graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system—without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

LIMITED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (b)(3) of the Rights in Technical Data -Noncommercial Items at DFARS 252.227-7013 (FEB 2014) and FAR 52.227-19 (DEC 2007).

Data contained herein pertains to a commercial product and/or commercial service (as defined in FAR 2.101) and is proprietary to NetApp, Inc. All NetApp technical data and computer software provided under this Agreement is commercial in nature and developed solely at private expense. The U.S. Government has a non-exclusive, non-transferrable, nonsublicensable, worldwide, limited irrevocable license to use the Data only in connection with and in support of the U.S. Government contract under which the Data was delivered. Except as provided herein, the Data may not be used, disclosed, reproduced, modified, performed, or displayed without the prior written approval of NetApp, Inc. United States Government license rights for the Department of Defense are limited to those rights identified in DFARS clause 252.227-7015(b) (FEB 2014).

Trademark information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.